

L7 Arrays

For Discord email - support@learnyard.com

Printing Arrays

Leetcode

→ Function

→ Code

→ Print

```
public static void print (int arr[])  
{
```

```
    for (int i = 0; i < arr.length; i++)
```

↳ size of array

```
        S.O. Print (arr[i]);
```

```
}
```

n
↓
arr[n]

Printing Arrays

1	2	3	4	5	6
0	1	2	3	4	5

size = 6
N

arr[N]

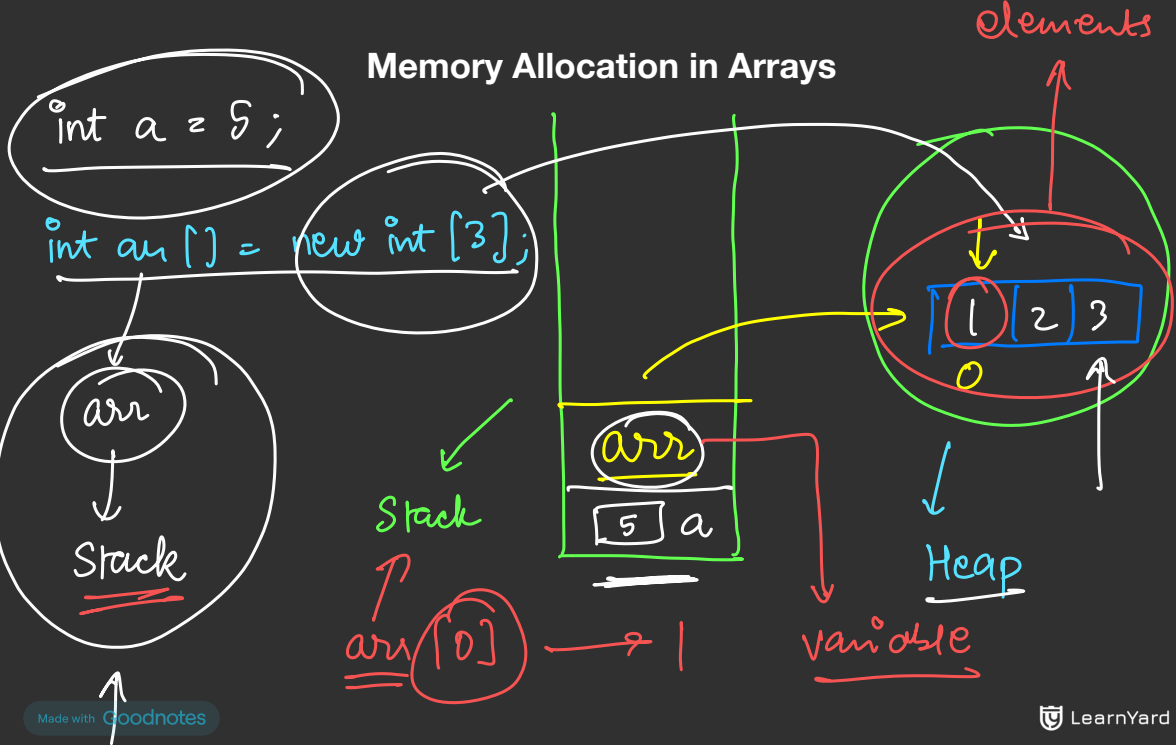
arr[5] = 6

0 - N - 1

arr[6]
↓

index out of bound

Memory Allocation in Arrays



✓ `int arr[] = new int[3];`

`arr[0] = 10;`

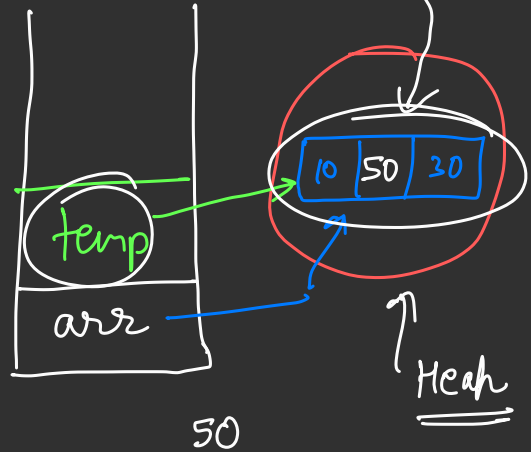
`arr[1] = 20;`

`arr[2] = 30;`

✓ `int temp[] = arr;`

↘ ~~new~~ ↘

`temp[1] = 50;`



↑
`S.O. Pen(arr[1])`

Basic Functions in array

arr.length → size of array

Arrays.sort (arr); → sort By default ascending

ascending 1 3 4 5 2 → arr

↑
order

Array.sort (arr);

→ arr

1 2 3 4 5

~~Pass by value~~ and Pass by reference in Array

```
public static void update (int arr[])  
{
```

```
    arr[0] = 500;
```

```
    arr[1] = 20;
```

```
}
```

Original value of
array will
change

✓ Question : Check if a element is present in array or not

└→ linear Search

Target = 4 → index = 3

→
1₀ 2₁ 3₂ 4₃ 5₄

answer = 1

↑
0
1

arr[i] == target

Question : Find maximum and minimum in a array

$\text{max} = 4$ ~~5~~ 7 $\rightarrow$ Final answer

1

3

4

5

2

7

6

6 > max

↑
i

max = arr[0]

Question : Check if array is sorted or not

$1 \leq 2 \leq 3 \leq 4 \leq 4 \leq 5 \leq 5 \leq 5$ ascending order

$1 \leq 4 \leq 3$ 5 6 7
false false not sorted

$arr[i] > arr[i+1] \rightarrow \text{false}$

Extra question [Missing number on Leetcode]

<https://leetcode.com/problems/missing-number/description/?ref=read.learnyard.com>

0 --- n $\rightarrow$ for loop (0 to n) }
 $\hookrightarrow$ formula $\rightarrow \frac{n * (n + 1)}{2}$

[3, 0, 1]
 $\uparrow$
0

sum = 0

sum = sum + arr[i]
 $\swarrow \quad \searrow$
3 1

sum = 3 + 1 = 4

Practice and Homework Questions

Homework questions -

<https://leetcode.com/problems/missing-number/description/?ref=read.learnyard.com>

Read these articles -

<https://read.learnyard.com/traversals-1/>

<https://read.learnyard.com/traversals-2/>

<https://read.learnyard.com/traversals-3>

<https://leetcode.com/problems/intersection-of-two-arrays/>

<https://leetcode.com/problems/two-sum-ii-input-array-is-sorted/>

Thank You!

Please practice more questions and examples as above !!